

Assignment 1.2: Calculating Comparable EMG Activity (2%)

Due: October 5th, 2026, at 11:59 p.m.

In the EMG sheet, **Column A** identifies the sleep–wake state for each row: Active Wake (AW), Quiet Wake (QW), NON-REM sleep (NR), or REM sleep (R). It also identifies the laser trials, which correspond to the EEG trials in Sheet 1 (“EEG”). Trial 1, for example, is the first time the laser was turned on.

***you can combine AW and QW as WAKE

Each row represents a 5-second epoch, and each trial lasts approximately 20 seconds. **Column B** contains the corresponding EMG values. These values have already been rectified, so no additional preprocessing is needed.

1) What do you need to do to be able to compare your EMG values between animals? Choose an appropriate method. You can use a method we discussed in the tutorial, but you are encouraged to explore other methods as well. Briefly explain your choice.

2) Usually, we additionally normalize EMG values by the average EMG value during a period of inactivity (EMG value per trial / average EMG value during inactivity). Which state would you consider a “period of inactivity”? Why would the other states be less suitable?

3) Once you have normalized your EMG data, calculate the average EMG value for each trial and add it to your EEG data frame. Your final data frame should look roughly like this:

**numbers are arbitrary

** EMG values may vary depending on the normalization method. Generally, you don’t need to normalize EMG values by their maximum value, so it’s okay if your results are above 100%. You can also express them as a ratio relative to your chosen state of inactivity.

Group Name	Power of Delta (%)	Power of Theta (%)	Power of Gamma (%)	Normalized EMG (%)
Trial 1	20	40	10	300
Trial 2	16	35	8	266
Trial 3	17	39	12	189

Keep in mind that you may have more frequencies than provided in this example table (see Assignment 1)